

Sayanjib Sur

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EDUCATION

Jadavpur University

Master of Engineering (M.E.) in Information Technology

- SGPA: 8.52

Kolkata, India

Sep 2025 – Present

Calcutta Institute of Engineering and Management

Bachelor of Technology in Information Technology

- SGPA: 8.46

Kolkata, India

Sep 2021 – June 2025

Ramakrishna Mission Boys' Home High School (H.S.), Rahara

12th Standard, Science

- WBCHSE: 88.4%

Kolkata, India

2021

Ramakrishna Mission Boys' Home High School (H.S.), Rahara

10th Standard

- WBBSE: 90.4%

Kolkata, India

2019

TECHNICAL SKILLS

Languages: Python, Java, C, SQL

Analytics & BI: Power BI, MS Excel, DAX, Power Query, DAX Studio

Machine Learning: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn

LLM & Agentic AI: LangChain, LangGraph, n8n

Web / ML Apps: Streamlit, JupyterLab

Tools: VS Code, Git, GitHub, MySQL Workbench

Cloud & APIs: Google Gemini API, REST APIs, Model Serving (Local/Cloud)

PROJECTS

SLM Benchmarking Pipeline for Structured Questionnaire Reasoning | *accepted at COMSYS 2026*

- Designed a cluster-aware prompting pipeline partitioning 5 SLMs (1.5B–7B) into context-length groups (2K–4K, 8K, 16K+) with distinct zero-shot, one-shot, and self-augmented prompt families calibrated per cluster.
- Conducted **10,800 inference runs per model** across 6 task types, 6 serialization formats (HTML, JSON, Markdown, Turtle, TXT, XML), and 5 diverse questionnaire datasets using local Ollama deployment on Google Colab Pro (A100).
- Introduced an MCQ-aware rescoring layer to recover valid option-letter predictions; best overall accuracy: **Vicuna-7B (35%)**; multi-hop inference and rule-based querying collapsed below **6%** across all models, exposing a hard reasoning ceiling in current SLMs.

SLM-Based Multi-Agent Frameworks for Mathematical Reasoning

[GitHub](#)

- Designed two complementary multi-agent methodologies evaluated on the **SVAMP benchmark** across 5 SLMs (1.5B–7B): **Pipeline Decomposition** (sequential Planner → Solver → Verifier with 3 controllable communication modes: natural, constrained, structured) and **Diverse Ensemble Reasoning** (parallel CoT, Formula-First, and Backward agents aggregated via majority vote).
- Best pipeline accuracy: **DeepSeek-R1-Distill-Qwen-1.5B (67%)** with natural communication; structured communication outperformed natural communication on verbose models by reducing noise entering the Solver; Planner-based distractor identification was the critical factor on SVAMP's deliberately misleading number sets.
- Diverse Ensemble achieved **LLaMA 3.2-3B at 81% accuracy** (vs. 77% CoT baseline), with a disagreement rate of 61% across strategies; introduced **NATE** (Normalized Accuracy per Thousand Tokens) as an efficiency metric for fair cross-model comparison under token-budget constraints.

NumFmt-SLM: Number Format Robustness Benchmark for SLMs

[GitHub](#)

- Built a reproducible benchmark testing whether mathematically equivalent number formats (plain, comma, words, scientific, abbreviated) change model predictions, evaluating **9 SLMs (0.6B–7B)** across **7 datasets** (5 MCQ + 2 generation) including AQuA-RAT, MMLU-Quant, GSM8K, and FinanceBench.
- Introduced **Format Sensitivity Score (FSS)** and **Format Flip Rate (FFR)** metrics with directional C→W/W→C flip tracking; found **18–86%** of MCQ answers flip under format changes alone, with FSS ranging from **0.042** (Qwen3-0.6B) to **0.095** (Phi-4-mini-IT).
- Showed format changes destroy correct answers $\sim 1.7\times$ more often than they create them (C→W 0.037 vs. W→C 0.022 mean); generation tasks exhibited **4–8 $\times$** higher FSS than MCQ tasks, with words-format numbers causing accuracy drops up to **16 pp**.

EXPERIENCE

Research Intern — IIT Patna (AI-ML-NLP Lab)

Feb 2026 – May 2026

- Working under Dr. Sriparna Saha on NLP for low-resource Indic languages; hands-on experience with multilingual models including Sarvam, Airavata, OpenHathi, IndicBART, Gemma, and LLaMA families.
- Built end-to-end pipelines to scrape, clean, and preprocess web data for low-resource languages, constructing a corpus of **15,000+ samples**.

Teacher & Founder — SS Classes

2023 – Present

- Founded a coaching initiative for **50+ undergrad students** (MAKAUT/RCCIT); taught Engineering Mathematics and Automata Theory, achieving **15–18 backlog clearances** and **15+ students with SGPA ≥ 8.0** .

Generative AI Developer Intern — AI Wallah

May 2025 – June 2025

- Built LLM-powered chatbots and a YouTube summarization tool using fine-tuning and Google Gemini API.

PUBLICATIONS

1. Prakash, D., Sur, S., and Saha, S. “A Multi-task Benchmark for Maithili and Bhojpuri NLP.” Submitted to *IEEE Access*. [Impact Factor: 4.2].
2. Sur, S., and Bhaumik, P. “Evaluating SLMs on Understanding Structured Frameworks using Cluster Aware Prompting.” *7th International Conference on Frontiers in Computing and Systems (COMSYS 2026)*. [Accepted].
3. Sur, S., Sil Sarma, A., and Singh, P. K. “NumFmt-SLM: Are Small Language Models Invariant to Number Surface Formats?” Submitted to *IEEE Calcutta Conference (CALCON 2026)*.
4. Tripathi, S., Kundu, S., Sur, S., and Singh, P. K. “Found but Not Used: A Full-Factorial Audit of RAG Distraction in Small Language Models.” Submitted to *IEEE Silchar Subsection Conference (SILCON 2026)*.
5. Sanbui, D., Bhol, S., Sur, S., and Singh, P. K. “Scaling Bias or Biasing Scale? Measuring Social Stereotype Bias Across Small Language Models.” Submitted to *IEEE Silchar Subsection Conference (SILCON 2026)*.
6. Sur, S., Sil Sarma, A., and Singh, P. K. “Structured Communication and Ensemble Diversity in SLM based Multi-Agent Systems: A Comparative Study.” Submitted to *International Conference for Convergence in Engineering (ICCE 2026)*.
7. Dash, S., Pahari, D., Sur, S., and Singh, P. K. “Small Models, Big Choices: A Systematic Study of Few-Shot Prompting versus Fine-Tuning for Tiny Language Models.” Submitted to *2026 IEEE 8th International Conference on Cybernetics, Cognition and Machine Learning Applications (ICCCMLA)*.
8. Sur, S., Jariwala, K. B., Kundu, A., Mehta, P. D., and Singha Deb, R. “Size Isn’t Safety: Prompt-Injection Robustness of Small On-Device Language Models.” Submitted to *International Conference on Machine Intelligence and Network Data Security (MINDS) 2026*.
9. Pahari, D., Sur, S., Rakshit, N., and Kundu, A. “Which Keypoints Matter? An Ablation Study for On-Device Indian Sign Language Recognition.” Submitted to *International Conference on Machine Intelligence and Network Data Security (MINDS) 2026*.
10. Sur, S., Ghosh, D., and Singh, P. K. “Small Language Models: A Survey of Architecture, Scaling, Applications, and Deployment (2022–2026).” Submitted to the Special Issue “Large Language Models for Cutting Edge Applications in Science and Humanities,” *Big Data and Cognitive Computing (BDCC)*, MDPI. [Impact Factor: 5.3].

LANGUAGES

Bengali, English, and Hindi

ACHIEVEMENTS

GATE CS/IT 2025 Qualified | Secured 96.27 percentile (Top 3.73% nationally)
UGC NET Dec 2025 Qualified (PhD Only) | Secured 97.45 percentile (Top 2.5% nationally)

LEADERSHIP / EXTRACURRICULAR

Celsius 2024 — Hackathon Co-Head 2024
Calcutta Institute of Engineering and Management

- Coordinated a hackathon with 20+ teams, managing logistics, planning, and execution.
- Ensured smooth scheduling, participant coordination, and competitive coding environment.